

Week 8 Report

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November 6, 2025

- 1 Single-cell gene set analysis Methods
- 2 scPS
- 3 experiment setting
- 4 Result

The commonly used scGSA methods are based on

- 1 Ranking
- 2 Aggregating gene expression

These methods have several problems:

- 1 sensitive to gene set size
- 2 sparsity
- 3 poor reproducibility

Existing Methods

The commonly used scGSA methods are based on

① Ranking-bases

- ① **ssGSEA** : calculated by the Kolmogorov-Smirnov-like random walk statistic
- ② **UCell** : calculates the Mann-Whitney U statistic
- ③ **AUCCell** : calculates the area under the curve (AUC) score among all ranked genes
- ④ **JASMINE** : calculates the enrichment of the signature using the mean rank of the expressed genes, the proportion of expressed to nonexpressed genes and the proportion of expressed genes in the gene set

Existing Methods

① Count-based

- ① **AddModuleScore** : a function in Seurat package and calculates scores for individual cells by aggregating the expression levels of each gene set, which are then subtracted by the aggregated expression of control feature sets randomly selected as background
- ② **SCSE** : calculated as a sum of gene expression within gene sets divided by the sum of gene expression for each cell

scPS

$$scPS = \mu \times \sqrt{\sum_{i=1}^m (s_i - s_{min}) \times v_i} \quad (1)$$

μ is the mean gene expression of the gene set.

s_i is the unweighted principal component score (PC).

s_{min} is the minimum of the s_i among all the cells.

v_i is the percentage of variance explained by the PC.

m is the number of PCs at which 50% cumulative variance is explained.

Result

scGSA methods for the comparative analysis

- ① Effect of the **cell count** on scGSA performance
- ② Effect of **gene set size** on scGSA performance
- ③ Effect of the **noise level** on scGSA performance
- ④ Effect of the **condition-specific genes** on scGSA performance.

zero inflation

Common "zero inflation" issues in single-cell RNA data:

- ① Many genes show an expression level of 0 (this could be due to genuine lack of expression or technical dropout);
- ② PCA may be dominated by these zero values, causing false signals;
- ③ Genes with low expression levels may be mistakenly identified as important variants.

To address this issue, paper used **scImpute**.

ScImpute effectively reduces dropout rates in scRNA-seq data by employing a sophisticated model that combines **gamma and normal distributions**.

Gene set simulation

- ① Gene set size : 10 500
- ② Noise : 0, 20, 50, 80, 100%
- ③ Generate 100 Gene set for each
- ④ Total 5,000 Gene set

Data simulation and signal assignment

Sample sizes of 20, 50, 200 and 500 cells in each group.

Data

- 1 Splatter simulated data (SSD) : generated using the Splatter package.
- 2 real-world simulated data (RWSD) were generated using log-transformed publicly available data.

Scenario

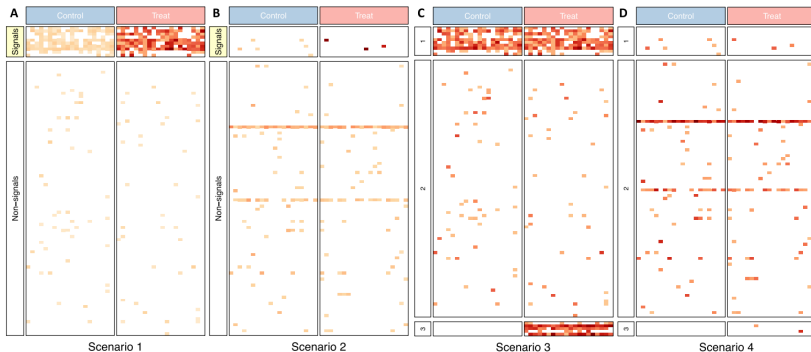
Gene expression (treatment (20%) > control)

- ① scenario 1 : At least 50 densely expressed genes (90% SSD vs 75% RWSD)
- ② scenario 2 : Random choose 550 genes

To evaluate the zero imputation effect

- ① no gene received any signal in these scenarios
- ② scenario 3 : introduced 250 (2%) more genes into the treatment group.
- ③ scenario 4 : introduced 250 (2%) more genes into the treatment group (sparse).

Scenario



Recovery rate

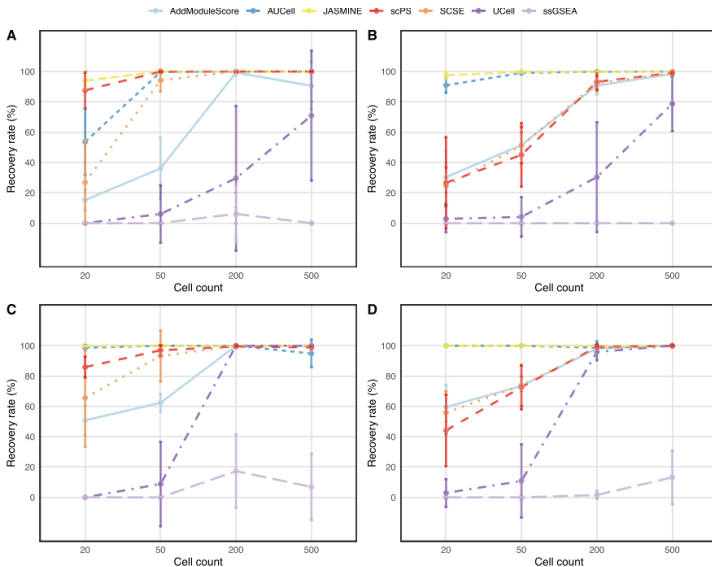
Recovery rate:

$$\frac{\text{Number of correctly identified gene sets}}{\text{Total number of simulated gene sets}} \quad (2)$$

Find Gene set

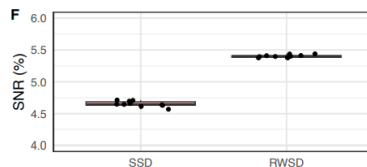
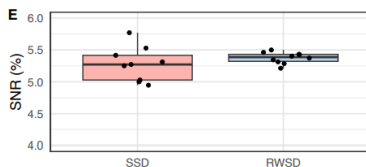
- Wilcoxon test
- FDR
- p-value < 0.05

Effect of the cell count on scGSA performance

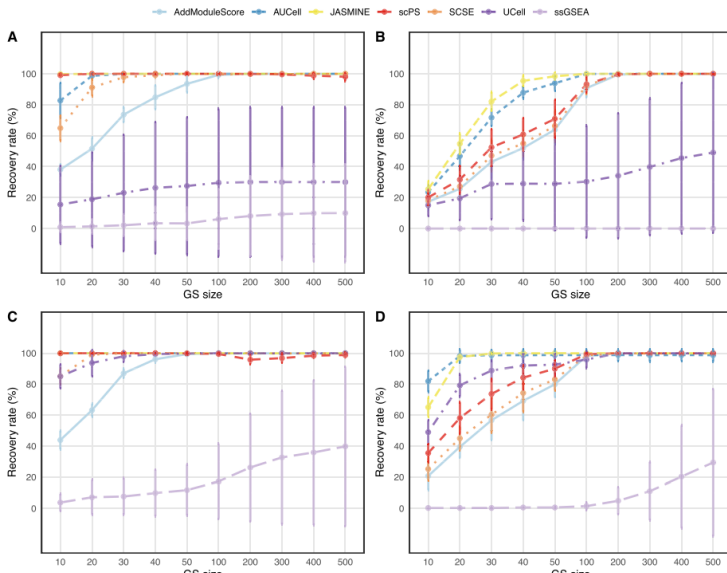


Effect of the cell count on scGSA performance

- 1 AddModuleScore, SCSE and UCell improved as the number of cells increased in both scenarios
- 2 UCell had a high standard deviation, attributable to the relatively high variability in the long tail of bottomranking genes.
- 3 The results in RWSD were better than those in SSD, which had a higher signal-to-non-signal ratios (SNR) and relatively lower sparsity.



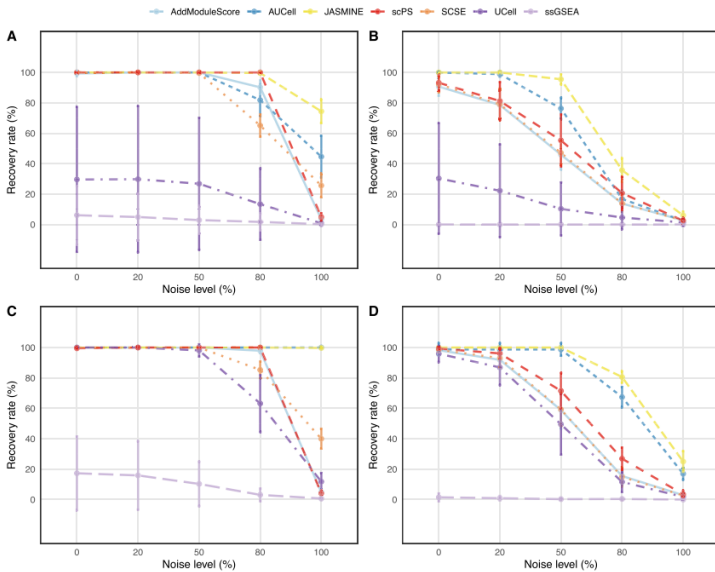
Effect of gene set size on scGSA performance



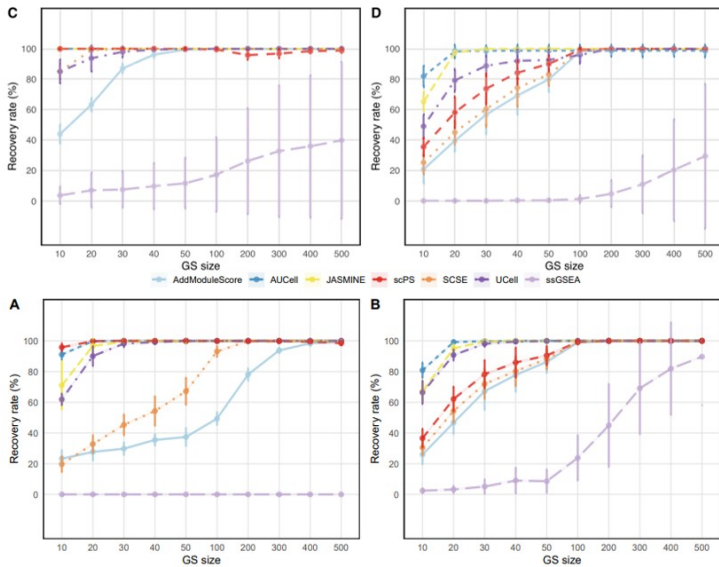
Effect of gene set size on scGSA performance

- 1 In scenario 1, 2, all differentially expressed gene sets between sizes of 200 and 500 were identified by all methods except ssGSEA
- 2 AddModuleScore performed well with a gene set size >50
- 3 UCell and ssGSEA performed poorly across all gene set sizes in SSD.
- 4 AUCell performed better when the data had a higher SNR and lower dropout rate upon zero imputation because AUCell randomly assigned ranks to genes with the same expression, leading to skewed results.
- 5 JASMINE considers the mean rank of the gene set and the ratio of expressed and unexpressed genes within or outside the gene set, which leads to a higher recovery rate when the data has a higher dropout rate, such as in single-cell datasets (SSD).

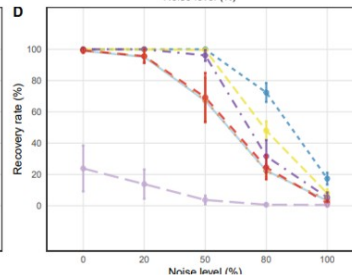
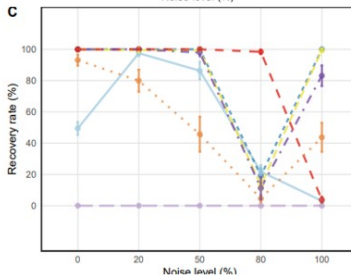
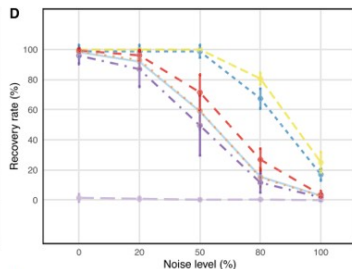
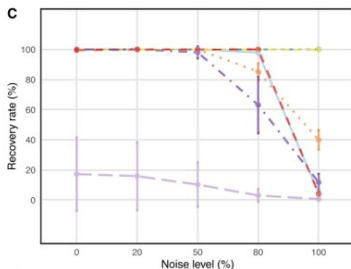
Effect of the noise level on scGSA performance



Zero-impute data



Zero-impute data



The sensitivity and false-positive rate of scGSA performance

Table 1. The sensitivity and false-positive rate of scGSA performance

	Sensitivity		False Positive Rate	
	SSD	RWSD	SSD	RWSD
Scenario 1				
JASMINE	100.0 ± 0.00	100.0 ± 0.00	74.4 ± 7.68	99.7 ± 0.48
AddModuleScore	99.3 ± 1.89	100.0 ± 0.00	3.5 ± 1.18	4.9 ± 1.66
AUCell	100.0 ± 0.00	100.0 ± 0.00	44.6 ± 13.66	100.0 ± 0.00
scPS	100.0 ± 0.00	99.5 ± 0.53	4.9 ± 2.47	4.1 ± 1.97
SCSE	100.0 ± 0.00	100.0 ± 0.00	25.6 ± 7.75	40.0 ± 6.51
UCell	29.6 ± 47.67	100.0 ± 0.00	0.8 ± 1.62	11.8 ± 5.73
ssGSEA	6.1 ± 19.29	17.3 ± 24.25	0.2 ± 0.63	0.7 ± 1.16
Scenario 2				
JASMINE	100.0 ± 0.00	100.0 ± 0.00	6.0 ± 2.45	25.0 ± 6.78
AddModuleScore	90.7 ± 6.04	98.2 ± 2.39	2.7 ± 1.70	2.9 ± 1.52
AUCell	99.9 ± 0.32	98.7 ± 4.11	2.5 ± 1.65	17.0 ± 3.89
scPS	93.2 ± 5.29	99.3 ± 0.82	2.5 ± 1.72	3.5 ± 2.32
SCSE	92.3 ± 5.54	98.6 ± 1.65	2.6 ± 1.26	2.7 ± 1.64
UCell	30.3 ± 36.25	95.9 ± 5.40	1.3 ± 2.26	1.8 ± 1.99
ssGSEA	0.0 ± 0.00	1.5 ± 2.59	0.0 ± 0.00	0.0 ± 0.00

Sensitivity is the recovery rate at a noise level of 0%, whereas false-positive rate is the recovery rate at a noise level of 100% with the gene set size of 100 and cell count of 200.

The sensitivity and false-positive rate of scGSA performance

- 1 This increased detection of false-positive gene sets by JASMINE and AUCell can be attributed to their reliance on gene expression rank without accounting for the magnitude of changes.
- 2 scPS and AddModuleScore return a few false-positive gene sets at a 100% noise level in data with and without zero imputation, which increases their accuracy and reliability.

Effect of the condition-specific genes

